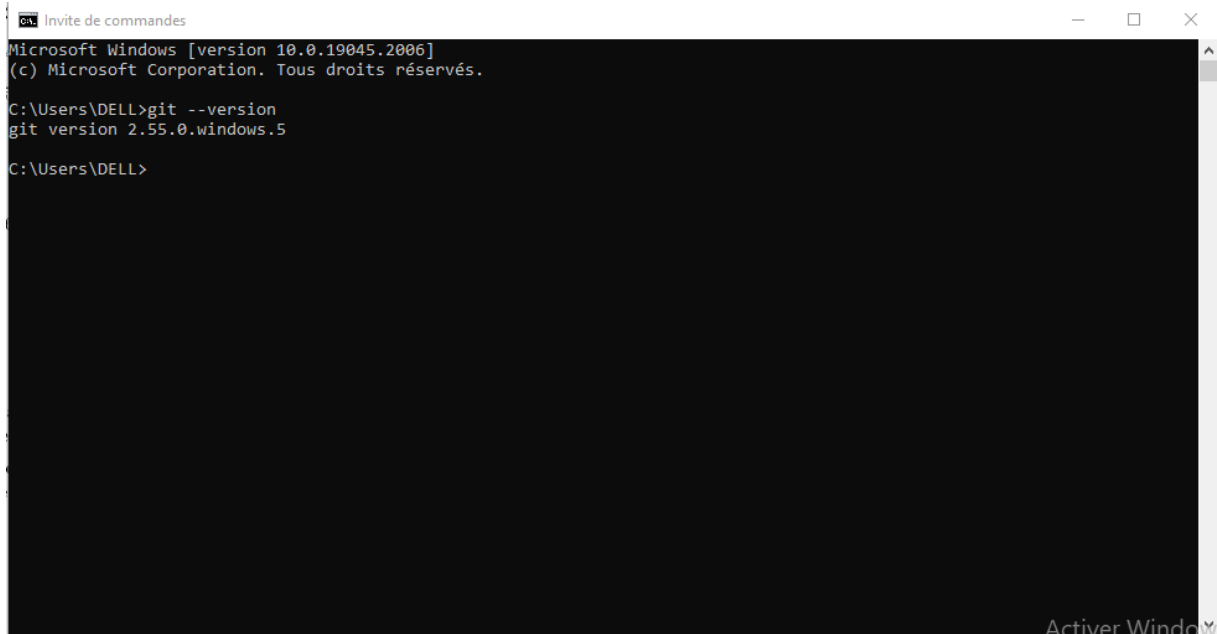


TP1 :introduction to the Git

1-git installation



```
Microsoft Windows [version 10.0.19045.2006]
(c) Microsoft Corporation. Tous droits réservés.

C:\Users\DELL>git --version
git version 2.55.0.windows.5

C:\Users\DELL>
```

2-repos creation

```
C:\Users\DELL\tpgit>git init
Initialized empty Git repository in C:/Users/DELL/tpgit/.git/

C:\Users\DELL\tpgit>dir /a .git
Le volume dans le lecteur C n'a pas de nom.
Le numéro de série du volume est E891-2FA3

Répertoire de C:\Users\DELL\tpgit\.git

17/09/2026  12:02    <DIR>        .
17/09/2026  12:02    <DIR>        ..
17/09/2026  12:02                130 config
17/09/2026  12:02                73 description
17/09/2026  12:02                23 HEAD
17/09/2026  12:02    <DIR>        hooks
17/09/2026  12:02    <DIR>        info
17/09/2026  12:02    <DIR>        objects
17/09/2026  12:02    <DIR>        refs
                3 fichier(s)                226 octets
                6 Rép(s) 49.582.997.504 octets libres

C:\Users\DELL\tpgit>
```

3-commits

```
C:\Users\DELL\tpgit>git log --graph --oneline --decorate --all
* f9166e9 (HEAD -> master) added kiwi to fruits
* 2163e6c added orange to fruits
* 49b912e added ananas to fruits
* e242934 added apple to fruits
```

```
C:\Users\DELL\tpgit>git diff fruits.txt

C:\Users\DELL\tpgit>git log
commit f9166e915a8758680d61bb90c5d5c3cedae07397 (HEAD -> master)
Author: hiba mhirit <hiba.mhirit@um6p.ma>
Date: Thu Sep 17 12:21:58 2026 +0200

    added kiwi to fruits

commit 2163e6cbbc1784b526b63067ea904d5021fbdd43
Author: hiba mhirit <hiba.mhirit@um6p.ma>
Date: Thu Sep 17 12:21:20 2026 +0200

    added orange to fruits

commit 49b912e2f522328edb7641a73c82ff12da43748e
Author: hiba mhirit <hiba.mhirit@um6p.ma>
Date: Thu Sep 17 12:19:34 2026 +0200

    added ananas to fruits

commit e242934df8b0fe4558e271f470f375229c782c61
Author: hiba mhirit <hiba.mhirit@um6p.ma>
Date: Thu Sep 17 12:14:44 2026 +0200

    added apple to fruits

C:\Users\DELL\tpgit>
```

4-branching

```
C:\Users\DELL\tpgit>git branch
  master
* vegetables

C:\Users\DELL\tpgit>dir
Le volume dans le lecteur C n'a pas de nom.
Le numéro de série du volume est E891-2FA3

Répertoire de C:\Users\DELL\tpgit

17/09/2026  12:35    <DIR>          .
17/09/2026  12:35    <DIR>          ..
17/09/2026  12:21             34 fruits.txt
17/09/2026  12:18              0 git
17/09/2026  12:35              0 sauces.txt
17/09/2026  12:34             27 vegetables.txt
               4 fichier(s)                61 octets
               2 Rép(s) 49.577.889.792 octets libres

C:\Users\DELL\tpgit>git add sauces.txt

C:\Users\DELL\tpgit>git status
On branch vegetables
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   sauces.txt

Untracked files:
  (use "git add <file>..." to include in what will be committed)
  git
```

```
C:\Users\DELL\tpgit>git branch
master
* sauces
vegetables

C:\Users\DELL\tpgit>git log
commit 09da92443dc63bb5734b073695d6ce3aa3fed6fd (HEAD -> sauces)
Author: hiba mhirit <hiba.mhirit@um6p.ma>
Date: Thu Sep 17 12:39:11 2026 +0200

    added mayonaise to sauces

commit a9c1161bafda64e83cd5373095bedb430cc1ace9
Author: hiba mhirit <hiba.mhirit@um6p.ma>
Date: Thu Sep 17 12:38:24 2026 +0200

    added ketchup to sauces

commit 8c040823d24f091c90e9a361c03b57114962db5d (vegetables)
Author: hiba mhirit <hiba.mhirit@um6p.ma>
Date: Thu Sep 17 12:34:49 2026 +0200

    added carrot to vegetables

commit e2940acea8f9b1dc94240bcb0e146a6a7b8e65b9
Author: hiba mhirit <hiba.mhirit@um6p.ma>
Date: Thu Sep 17 12:34:00 2026 +0200

    added tomato to vegetables

commit 80852b111072cf74ddd809917486c8b0d3325110
Author: hiba mhirit <hiba.mhirit@um6p.ma>
Date: Thu Sep 17 12:33:26 2026 +0200

    added potato to vegetables

commit f9166e915a8758680d61bb90c5d5c3cedae07397 (master)
Author: hiba mhirit <hiba.mhirit@um6p.ma>
Date: Thu Sep 17 12:21:58 2026 +0200

    added kiwi to fruits
```

5-graph

The screenshot shows the tpgit application interface. At the top, the title bar reads "tpgit: --all - gitk". Below the title bar is a menu bar with "File", "Edit", "View", and "Help". The main area is divided into three panes. The left pane shows a commit graph with a vertical list of commits. The middle pane shows a diff view for the commit "added mayonaise to sauces". The right pane shows a list of commits with their authors and timestamps.

Commit graph (left pane):

- master sauces added mayonaise to sauces
- added ketchup to sauces
- vegetables added carrot to vegetables
- added tomato to vegetables
- added potato to vegetables
- added kiwi to fruits
- added orange to fruits
- added ananas to fruits
- added apple to fruits

Diff view (middle pane):

Commit ID: 09da92443dc63bb5734b073695d6ce3aa3fed6fd

Find: commit containing: []

Search: []

Diff view content:

```
Author: hiba mhirit <hiba.mhirit@um6p.ma> 2026-09-17 12:39:11
Committer: hiba mhirit <hiba.mhirit@um6p.ma> 2026-09-17 12:39:11
Parent: a9c1161bafda64e83cd5373095bedb430cc1ace9 (added ketchup to sauces)
Branches: master, sauces
Follows:
Precedes:

    added mayonaise to  sauces

----- sauces.txt -----
index f755ae2..6e32db7 100644
@@ -1 +1,2 @@
 ketchup
+mayonaise
```

Commit list (right pane):

Commit	Author	Timestamp
added mayonaise to sauces	hiba mhirit <hiba.mhirit@um6p.ma>	2026-09-17 12:39:11
added ketchup to sauces	hiba mhirit <hiba.mhirit@um6p.ma>	2026-09-17 12:38:24
added carrot to vegetables	hiba mhirit <hiba.mhirit@um6p.ma>	2026-09-17 12:34:49
added tomato to vegetables	hiba mhirit <hiba.mhirit@um6p.ma>	2026-09-17 12:34:00
added potato to vegetables	hiba mhirit <hiba.mhirit@um6p.ma>	2026-09-17 12:33:26
added kiwi to fruits	hiba mhirit <hiba.mhirit@um6p.ma>	2026-09-17 12:21:58
added orange to fruits	hiba mhirit <hiba.mhirit@um6p.ma>	2026-09-17 12:21:20
added ananas to fruits	hiba mhirit <hiba.mhirit@um6p.ma>	2026-09-17 12:19:34
added apple to fruits	hiba mhirit <hiba.mhirit@um6p.ma>	2026-09-17 12:14:44

6-github config

The screenshot shows a GitHub authentication success message. At the top, there is a logo consisting of the GitHub Octocat icon, a plus sign, and the GitLab logo. Below the logo, the text "Authentication Succeeded" is displayed in a large, bold font. Underneath, a smaller line of text says "You may now close this tab and return to the application."

Authentication Succeeded

You may now close this tab and return to the application.

```
C:\Users\DELL\tpgit>git clone https://github.com/SalmaMana/tpgit.git tp-git-binome
Cloning into 'tp-git-binome'...
remote: Enumerating objects: 12, done.
remote: Counting objects: 100% (12/12), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 12 (delta 1), reused 12 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (12/12), done.
Resolving deltas: 100% (1/1), done.

C:\Users\DELL\tpgit>
```

7-collaborative work

```
C:\Users\DELL\tpgit>cd tp-git-binome

C:\Users\DELL\tpgit\tp-git-binome>git log --graph --oneline --decorate --all
* 6cba6bc (HEAD -> master, origin/master, origin/HEAD) Add vanilla sauce
* a9fea63 Add tomato
* 86d2aba Add carrot
* 9347f57 orange

C:\Users\DELL\tpgit\tp-git-binome>
```

7-1 merging herbs with master

```
C:\Users\DELL\tpgit\tp-git-binome>git log --oneline --all
1cec54e (HEAD -> master, herbs) added dill to herbs
043d29d added chives to herbs
6cba6bc (origin/master, origin/HEAD) Add vanilla sauce
a9fea63 Add tomato
86d2aba Add carrot
9347f57 orange

C:\Users\DELL\tpgit\tp-git-binome>git branch
  herbs
* master

C:\Users\DELL\tpgit\tp-git-binome>
```

Activer Windows

7-2 checking if we have both spices and herbs

After checking we found that the ids are the same:

```

C:\Users\DELL\tpgit\tp-git-binome>git log --oneline --graph --all
* 96dbd99 (origin/herbs, herbs) Add basil to herbs
| * 7dd01ba (refs/stash) On master: save current files before fixing master
|/|\
| * 360c74b untracked files on master: 1cec54e added dill to herbs
| * 3387a23 index on master: 1cec54e added dill to herbs
|/
* 1cec54e (backup-old-master) added dill to herbs
* 043d29d added chives to herbs
* 6cba6bc Add vanilla sauce
* a9fea63 Add tomato
* 86d2aba Add carrot
* 9347f57 orange
* a0952f2 (HEAD -> master, origin/master, origin/HEAD) Add salt to spices
* 6670cc4 Add oregano to spices
* d10bcc1 Add Béchamel to sauces
* 0204298 Add Pesto to sauces
* b4336d1 Add zucchini to vegetables
* dd4c52b Add eggplant to vegetables
* a6a2360 Add carrots
* 9f89ba9 Add cherries
* c853f12 Add Mango
* e3d3238 Add apple
* ddeeed2 Add banana
* 364702d Add orange

C:\Users\DELL\tpgit\tp-git-binome>

```

10-merge with conflicts

1-conflict 1

My change(hiba mherit) :

```

C:\Users\DELL\tpgit\tp-git-binome>powershell -Command "(Get-Content fruits.txt) -replace 'watermelon','anas' | Set-content fruits.txt"
C:\Users\DELL\tpgit\tp-git-binome>type fruits.txt
orange
anas
apple
mango
cherries
C:\Users\DELL\tpgit\tp-git-binome>git add fruits.txt
C:\Users\DELL\tpgit\tp-git-binome>git commit -m "Change watermelon to ananas"
[master 9476773] Change watermelon to ananas
1 file changed, 1 insertion(+), 1 deletion(-)
C:\Users\DELL\tpgit\tp-git-binome>

```

Activer Windows
Accédez aux paramètres

```
C:\Users\DELL\tpgit\tp-git-binome>git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 377 bytes | 377.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/SalmaMana/tpgit.git
   f03306b..9476773  master -> master

C:\Users\DELL\tpgit\tp-git-binome>
```

My collaborator's change(salma mana):

```
SALMA.PC@DESKTOP-U2NA2K6 MINGW64 ~/Documents/tpgit (master)
$ cat fruits.txt
orange
watermelon
apple
mango
cherrys

SALMA.PC@DESKTOP-U2NA2K6 MINGW64 ~/Documents/tpgit (master)
$ sed -i 's/watermelon/strawberry/' fruits.txt
bash: sed -i s/watermelon/strawberry/fruits.txt: No such file or directory

SALMA.PC@DESKTOP-U2NA2K6 MINGW64 ~/Documents/tpgit (master)
$ sed -i 's/watermelon/starawberry/' fruits.txt

SALMA.PC@DESKTOP-U2NA2K6 MINGW64 ~/Documents/tpgit (master)
$ cat fruits.txt
orange
starawberry
apple
mango
cherrys

SALMA.PC@DESKTOP-U2NA2K6 MINGW64 ~/Documents/tpgit (master)
$ git add fruits.txt
warning: in the working copy of 'fruits.txt', LF will be replaced by CRLF the next time Git touches it

SALMA.PC@DESKTOP-U2NA2K6 MINGW64 ~/Documents/tpgit (master)
$ git commit -m "change watermelon to strawberry"
[master 198089b] change watermelon to strawberry
1 file changed, 1 insertion(+), 1 deletion(-)

SALMA.PC@DESKTOP-U2NA2K6 MINGW64 ~/Documents/tpgit (master)
$ git push
To https://github.com/SalmaMana/tpgit.git
 ! [rejected]        master -> master (fetch first)
error: failed to push some refs to 'https://github.com/SalmaMana/tpgit.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.

SALMA.PC@DESKTOP-U2NA2K6 MINGW64 ~/Documents/tpgit (master)
$
```

Conflict resolution :

My work :

```

C:\Users\DELL\tpgit\tp-git-binome>git pull
remote: Enumerating objects: 8, done.
remote: Counting objects: 100% (8/8), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 4 (delta 0), reused 4 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (4/4), 568 bytes | 21.00 KiB/s, done.
From https://github.com/SalmaMana/tpgit
   9476773..1ebc913  master    -> origin/master
Updating 9476773..1ebc913
Fast-forward

C:\Users\DELL\tpgit\tp-git-binome>type fruits.txt
orange
ananas
apple
mango
cherrys

C:\Users\DELL\tpgit\tp-git-binome>git --no-pager log --oneline --graph --all
*   1ebc913 (HEAD -> master, origin/master, origin/HEAD) resolve merge conflict in fruits.txt
| \
| * 9476773 Change watermelon to ananas
| * 198089b change watermelon to strawberry
| /
* f03306b Change banana to watermelon
* a0952f2 Add salt to spices
* 6670cc4 Add oregano to spices
* d10bcc1 Add Béchamel to sauces
* 0204298 Add Pesto to sauces
* b4336d1 Add zucchini to vegetables
* dd4c52b Add eggplant to vegetables
* a6a2360 Add carrots
* 9f89ba9 Add cherrys
* c853f12 Add Mango
* e3d3238 Add apple
* ddeeed2 Add banana
* 364702d Add orange
* 96dbd99 (origin/herbs, herbs) Add basil to herbs
| * 7dd01ba (refs/stash) On master: save current files before fixing master
| / \
| * 360c74b untracked files on master: 1cec54e added dill to herbs
| * 3387a23 index on master: 1cec54e added dill to herbs
| /

```

Her work :

```
SALMA.PC@DESKTOP-U2NA2K6 MINGW64 ~/Documents/tpgit (master|MERGING)
$ git checkout --theirs fruits.txt
Updated 1 path from the index

SALMA.PC@DESKTOP-U2NA2K6 MINGW64 ~/Documents/tpgit (master|MERGING)
$ cat fruits.txt
orange
ananas
apple
mango
cherrys

SALMA.PC@DESKTOP-U2NA2K6 MINGW64 ~/Documents/tpgit (master|MERGING)
$ git add fruits.txt

SALMA.PC@DESKTOP-U2NA2K6 MINGW64 ~/Documents/tpgit (master|MERGING)
$ git commit -m "resolve merge conflict in fruits.txt"
[master 1ebc913] resolve merge conflict in fruits.txt

SALMA.PC@DESKTOP-U2NA2K6 MINGW64 ~/Documents/tpgit (master)
$ git push
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 588 bytes | 588.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/SalmaMana/tpgit.git
 9476773..1ebc913  master -> master
```

2-conflict 2

My work :

```

C:\Users\DELL\tpgit\tp-git-binome>git log --oneline --graph --all
* a7926e9 (HEAD -> master, origin/master, origin/HEAD) Resolve conflict in vegetables.txt
| \
| * 328d3ce change carrot to potato in vegetables.txt
| * | 24163b8 Change carrots to tomato
| /
| * 1ebc913 resolve merge conflict in fruits.txt
| \
| * 9476773 Change watermelon to ananas
| * | 198089b change watermelon to strawberry
| /
| * f03306b Change banana to watermelon
| * a0952f2 Add salt to spices
| * 6670cc4 Add oregano to spices
| * d10bcc1 Add Béchamel to sauces
| * 0204298 Add Pesto to sauces
| * b4336d1 Add zucchini to vegetables
| * dd4c52b Add eggplant to vegetables
| * a6a2360 Add carrots
| * 9f89ba9 Add cherries
| * c853f12 Add Mango
| * e3d3238 Add apple
| * ddeed2 Add banana
| * 364702d Add orange
| * 96dbd99 (origin/herbs, herbs) Add basil to herbs
| * 7dd01ba (refs/stash) On master: save current files before fixing master
| / \
| | * 360c74b untracked files on master: 1cec54e added dill to herbs
| | * 3387a23 index on master: 1cec54e added dill to herbs
| /
| * 1cec54e (backup-old-master) added dill to herbs
| * 043d29d added chives to herbs
| * 6cba6bc Add vanilla sauce
| * a9fea63 Add tomato
| * 86d2aba Add carrot
| * 9347f57 orange

```

Salma's work :

```

SALMA.PC@DESKTOP-U2NA2K6 MINGW64 ~/Documents/tpgit (master)
$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 302 bytes | 151.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/SalmaMana/tpgit.git
   1ebc913..328d3ce  master -> master

SALMA.PC@DESKTOP-U2NA2K6 MINGW64 ~/Documents/tpgit (master)
$ git pull
remote: Enumerating objects: 10, done.
remote: Counting objects: 100% (10/10), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 6 (delta 2), reused 6 (delta 2), pack-reused 0 (from 0)
Unpacking objects: 100% (6/6), 614 bytes | 38.00 KiB/s, done.
From https://github.com/SalmaMana/tpgit
   328d3ce..a7926e9  master      -> origin/master
Updating 328d3ce..a7926e9
Fast-forward
   vegetables.txt | 4 ++++
   1 file changed, 4 insertions(+)

SALMA.PC@DESKTOP-U2NA2K6 MINGW64 ~/Documents/tpgit (master)
$ cat vegetables.txt
<<<<<<< HEAD
tomato
=====
potato
>>>>>>> 328d3ceeece357c5817c8ceab48fe9308419c3ac
eggplant
zucchini

SALMA.PC@DESKTOP-U2NA2K6 MINGW64 ~/Documents/tpgit (master)
$ git log --oneline --graph --all
*   a7926e9 (HEAD -> master, origin/master, origin/HEAD) Resolve conflict in vegetables.txt
| \
| * 328d3ce change carrot to potato in vegetables.txt
* | 24163b8 Change carrots to tomato
| /
*   1ebc913 resolve merge conflict in fruits.txt
| \
| * 9476773 Change watermelon to ananas
* | 198089b change watermelon to strawberry
| /
* f03306b Change banana to watermelon
* a0952f2 (spices) Add salt to spices
* 6670cc4 Add oregano to spices
* d10bcc1 (vegetables, sauces) Add Béchamel to sauces
* 0204298 Add Pesto to sauces
* b4336d1 Add zucchini to vegetables
* dd4c52b Add eggplant to vegetables
* a6a2360 Add carrots
* 9f89ba9 Add cherries
* c853f12 Add Mango
* e3d3238 Add apple
* ddeeed2 Add banana
* 364702d Add orange
* 96dbd99 (origin/herbs, herbs) Add basil to herbs
* 1cec54e added dill to herbs
* 043d29d added chives to herbs
* 6cba6bc Add vanilla sauce
* a9fea63 Add tomato
* 86d2aba Add carrot
* 9347f57 orange

```

Author : Hiba mhirit

collaborator : salma mana

